REPORT

Reviewer report: pytoolz/toolz#665: Fix pluck default

handling when indexing raises TypeError

Run toolz-665, tested 2026-11-14 12:51 IST by ops/daily_run.py (route source-pair).
Proof-of-Change commit

ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-14 12:58 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

VERIFIED

Recommended action: Approve with notes Confidence: High

The PR's tests behave the same or better than on main, in all 3 runs per side. Every check passed and every

claim cites a log line.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: Fix pluck default handling when indexing raises TypeError (pytoolz/toolz#665).

• What we observed: 1 test(s) fail on main and pass on the PR; 79 test(s) ran on main, 80 on the PR; 1 new

test(s) pass, 78 unchanged; probe: claim(s) C1, C2, C4, C5, C6 proven.

• What you should do: You can approve; read the notes under 'What went wrong' first.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 10.6 GB (of 15.6 GB).

• Setup check: free disk 92.4 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: toolz python setup install instructions https://github.com/pytoolz/toolz site:github.com/pytoolz/toolz

• Query: toolz python documentation install https://github.com/pytoolz/toolz site:github.com/pytoolz/toolz

• Query: toolz python install requirements https://pypi.org/project/toolz/ https://github.com/pytoolz/toolz

site:pypi.org

• Source: toolz/README.rst at master · pytoolz/toolz · GitHub —

https://github.com/pytoolz/toolz/blob/master/README.rst

• Source: toolz/README.rst at master · pytoolz/toolz · GitHub —

http://github.com/pytoolz/toolz/blob/master/README.rst

• Source: Toolz — https://github.com/pytoolz/toolz/raw/refs/heads/master/README.rst

• Source: toolz/README.rst at 451af60dec590a6010e2babdbf391ea8f815122f · pytoolz/toolz · GitHub —

https://github.com/pytoolz/toolz/blob/451af60dec590a6010e2babdbf391ea8f815122f/README.rst

• Source: toolz — https://pypi.org/project/toolz/

• Setup plan: the same packages on main and pr.

• Setup plan: install coverage (listed in the test extra) on main and pr.

• Setup plan: install pycodestyle (listed in the test extra) on main and pr.

• Setup plan: install pytest (listed in the test extra) on main and pr.

• Setup plan: install sphinx (listed in requirements.txt) on main and pr.

• Setup plan: install furo (listed in requirements.txt) on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.9

• Installed coverage 7.17.3 from pyproject (listed in the test extra) on main and pr.

• Installed pycodestyle 2.15.0 from pyproject (listed in the test extra) on main and pr.

• Installed pytest 9.2.0 from pyproject (listed in the test extra) on main and pr.

• Installed sphinx 9.1.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed furo 2025.12.19 from requirements.txt (listed in requirements.txt) on main and pr.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| C2 | Existing behavior is kept: tests that pass on main still pass (implied by every PR) | proven | no test that

passes on main fails on the PR (78 unchanged, 3 runs per side) | pytest-pr-1.log:8 «80 passed in 0.99s»

| P-C1 | pluck() raises TypeError when a single-index lookup fails, even when a default is supplied. (PR

description) | proven | probe model-probe1, c1.exception: value: "TypeError" -> null |

probe-model-probe1-pr-1.jsonl:1 «"observation": "c1.exception"»

| P-C2 | list(pluck("a", [[1, 2]], default=None)) previously raised TypeError. (PR description) | proven | probe

model-probe1, c2.result: value: null -> [null] | probe-model-probe1-pr-1.jsonl:4 «"observation": "c2.result"»

| P-C4 | get() returns the default for the same input. (PR description) | proven | probe model-probe1,

c4.get_exception: same on both sides: null | probe-model-probe1-pr-1.jsonl:5 «"observation":

"c4.get_exception"»

| P-C6 | The reported case fails before the fix and passes afterward. (PR description) | proven | probe

model-probe1, c6.outcome: value: "error" -> "pass" | probe-model-probe1-pr-1.jsonl:7 «"observation":

"c6.outcome"»

| P-C5 | Preserve lazy evaluation, multi-index behavior, and exception propagation when no default is supplied.

(PR description) | proven | probe model-probe2, c5.exception_when_no_default: same on both sides: null |

probe-model-probe2-pr-1.jsonl:1 «"observation": "c5.exception_when_no_default"»

Citation check (PASS): 6 citation(s), every one re-read from its log file and matched.

What changed in behavior

| What | Count | Examples

| Tests that actually ran (main / PR, fewest of the runs) | 79 / 80 | | Tests unchanged | 78 | | New tests that pass | 1 | -

| Tests that pass on main but fail on the PR | 0 | | Tests that are new in the PR and fail | 0 | | Tests that ran on main but are missing on the PR side | 0 | | Tests that changed result in another way (for example passed -> skipped) | 0 | | Tests that gave different results between the 3 runs of the same side (flaky) | 0 | | Tests that fail on main and pass on the PR | 1 | toolz.tests.test_itertoolz::test_pluck_default_typeerror

| Tests that are new or changed but did not run because an optional module is missing | 0 | | Tests that were re-parametrized (same test, new parameter ids; not counted as removed) | 0 | | Test modules that fail to load on the PR side only (their tests no longer run) | 0 | | Test modules that fail to load on main only (no baseline for their tests) | 0 | | Test modules that fail to load on both sides (their tests did not run) | 0 | | Tests that fail identically on both sides (already broken on main) | 0 | -

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs

per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the

PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked

against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model

fixtures are known-answer cases.

model-probe1 (model (nvidia/nemotron-3-super-120b-a12b, live), model-probes/model-probe1.py, frozen

82d552a3b38a at 2026-11-14T07:22:27.733556+00:00): Test single-index failure with default (C1, C2, C4, C6)

• P-C1 (PR description): pluck() raises TypeError when a single-index lookup fails, even when a default is

supplied. Expects the covered observations to differ.

• P-C2 (PR description): list(pluck("a", [[1, 2]], default=None)) previously raised TypeError. Expects the covered

observations to differ.

• P-C4 (PR description): get() returns the default for the same input. Expects the covered observations to stay the

same.

• P-C6 (PR description): The reported case fails before the fix and passes afterward. Expects the covered

observations to differ.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| c1.exception | P-C1 | as claimed | value: "TypeError" -> null | probe-model-probe1-main-1.jsonl:1

«"observation": "c1.exception"»
probe-model-probe1-pr-1.jsonl:1 «"observation": "c1.exception"»

| c1.result | P-C1 | as claimed | value: null -> [null] | probe-model-probe1-main-1.jsonl:2 «"observation":

"c1.result"»
probe-model-probe1-pr-1.jsonl:2 «"observation": "c1.result"»

| c1.result_type | P-C1 | as claimed | value: null -> "list" | probe-model-probe1-main-1.jsonl:3 «"observation":

"c1.result_type"»
probe-model-probe1-pr-1.jsonl:3 «"observation": "c1.result_type"»

| c2.result | P-C2 | as claimed | value: null -> [null] | probe-model-probe1-main-1.jsonl:4 «"observation":

"c2.result"»
probe-model-probe1-pr-1.jsonl:4 «"observation": "c2.result"»

| c4.get_exception | P-C4 | as claimed | same on both sides: null | probe-model-probe1-main-1.jsonl:5

«"observation": "c4.get_exception"»
probe-model-probe1-pr-1.jsonl:5 «"observation": "c4.get_exception"»

| c4.get_result | P-C4 | as claimed | same on both sides: null | probe-model-probe1-main-1.jsonl:6 «"observation":

"c4.get_result"»
probe-model-probe1-pr-1.jsonl:6 «"observation": "c4.get_result"»

| c6.outcome | P-C6 | as claimed | value: "error" -> "pass" | probe-model-probe1-main-1.jsonl:7 «"observation":

"c6.outcome"»
probe-model-probe1-pr-1.jsonl:7 «"observation": "c6.outcome"»

model-probe2 (model (nvidia/nemotron-3-super-120b-a12b, live), model-probes/model-probe2.py, frozen

708067df5bf6 at 2026-11-14T07:22:27.733681+00:00): Test lazy evaluation, multi-index behavior, and exception

propagation when no default (C5)

• P-C5 (PR description): Preserve lazy evaluation, multi-index behavior, and exception propagation when no

default is supplied. Expects the covered observations to stay the same.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| c5.exception_when_no_default | P-C5 | as claimed | same on both sides: null |

probe-model-probe2-main-1.jsonl:1 «"observation":

"c5.exception_when_no_default"»
probe-model-probe2-pr-1.jsonl:1 «"observation":

"c5.exception_when_no_default"»

| c5.lazy_type | P-C5 | as claimed | same on both sides: "map" | probe-model-probe2-main-1.jsonl:2

«"observation": "c5.lazy_type"»
probe-model-probe2-pr-1.jsonl:2 «"observation": "c5.lazy_type"»

| c5.multi_result | P-C5 | as claimed | same on both sides: [[1, 2], [3, 4]] | probe-model-probe2-main-1.jsonl:3

«"observation": "c5.multi_result"»
probe-model-probe2-pr-1.jsonl:3 «"observation": "c5.multi_result"»

diff-api (diff (auto-derived from base-side code, model-free); informational under baseline full,

diff-probes/diff-api.py, frozen 474281c1d3bc at 2026-11-14T07:22:27.736281+00:00): Public functions touched

by the PR, called with type-appropriate inputs on both sides.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| toolz.itertoolz.getter.input0 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(1)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:1 «"observation":

"toolz.itertoolz.getter.input0"»
probe-diff-api-pr-1.jsonl:1 «"observation": "toolz.itertoolz.getter.input0"»

| toolz.itertoolz.getter.input1 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(0)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:2 «"observation":

"toolz.itertoolz.getter.input1"»
probe-diff-api-pr-1.jsonl:2 «"observation": "toolz.itertoolz.getter.input1"»

| toolz.itertoolz.getter.input10 | - | same on both sides | same on both sides: {"repr": "<function

getter.<locals>.<lambda>>", "type": "function"} | probe-diff-api-main-1.jsonl:3 «"observation":

"toolz.itertoolz.getter.input10"»
probe-diff-api-pr-1.jsonl:3 «"observation": "toolz.itertoolz.getter.input10"»

| toolz.itertoolz.getter.input11 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter({})",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:4 «"observation":

"toolz.itertoolz.getter.input11"»
probe-diff-api-pr-1.jsonl:4 «"observation": "toolz.itertoolz.getter.input11"»

| toolz.itertoolz.getter.input2 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(-1)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:5 «"observation":

"toolz.itertoolz.getter.input2"»
probe-diff-api-pr-1.jsonl:5 «"observation": "toolz.itertoolz.getter.input2"»

| toolz.itertoolz.getter.input3 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(2.5)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:6 «"observation":

"toolz.itertoolz.getter.input3"»
probe-diff-api-pr-1.jsonl:6 «"observation": "toolz.itertoolz.getter.input3"»

| toolz.itertoolz.getter.input4 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter('abc')",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:7 «"observation":

"toolz.itertoolz.getter.input4"»
probe-diff-api-pr-1.jsonl:7 «"observation": "toolz.itertoolz.getter.input4"»

| toolz.itertoolz.getter.input5 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(1, 2, 3)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:8 «"observation":

"toolz.itertoolz.getter.input5"»
probe-diff-api-pr-1.jsonl:8 «"observation": "toolz.itertoolz.getter.input5"»

| toolz.itertoolz.getter.input6 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(None)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:9 «"observation":

"toolz.itertoolz.getter.input6"»
probe-diff-api-pr-1.jsonl:9 «"observation": "toolz.itertoolz.getter.input6"»

| toolz.itertoolz.getter.input7 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(True)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:10 «"observation":

"toolz.itertoolz.getter.input7"»
probe-diff-api-pr-1.jsonl:10 «"observation": "toolz.itertoolz.getter.input7"»

| toolz.itertoolz.getter.input8 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(False)",

"type": "itemgetter"} | probe-diff-api-main-1.jsonl:11 «"observation":

"toolz.itertoolz.getter.input8"»
probe-diff-api-pr-1.jsonl:11 «"observation": "toolz.itertoolz.getter.input8"»

| toolz.itertoolz.getter.input9 | - | same on both sides | same on both sides: {"repr": "operator.itemgetter(')", "type":

"itemgetter"} | probe-diff-api-main-1.jsonl:12 «"observation":

"toolz.itertoolz.getter.input9"»
probe-diff-api-pr-1.jsonl:12 «"observation":
"toolz.itertoolz.getter.input9"»

| toolz.itertoolz.pluck.input0 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:13 «"observation":
"toolz.itertoolz.pluck.input0"»
probe-diff-api-pr-1.jsonl:13

«"observation": "toolz.itertoolz.pluck.input0"»

| toolz.itertoolz.pluck.input1 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:14 «"observation":
"toolz.itertoolz.pluck.input1"»
probe-diff-api-pr-1.jsonl:14

«"observation": "toolz.itertoolz.pluck.input1"»

| toolz.itertoolz.pluck.input10 | - | same on both sides | same on both sides: {"repr": "<map object>",
"type":

"map"} | probe-diff-api-main-1.jsonl:15 «"observation":

"toolz.itertoolz.pluck.input10"»
probe-diff-api-pr-1.jsonl:15 «"observation":
"toolz.itertoolz.pluck.input10"»

| toolz.itertoolz.pluck.input11 | - | same on both sides | same on both sides: {"repr": "<map object>",
"type":

"map"} | probe-diff-api-main-1.jsonl:16 «"observation":

"toolz.itertoolz.pluck.input11"»
probe-diff-api-pr-1.jsonl:16 «"observation":
"toolz.itertoolz.pluck.input11"»

| toolz.itertoolz.pluck.input2 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:17 «"observation":
"toolz.itertoolz.pluck.input2"»
probe-diff-api-pr-1.jsonl:17

«"observation": "toolz.itertoolz.pluck.input2"»

| toolz.itertoolz.pluck.input3 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:18 «"observation":
"toolz.itertoolz.pluck.input3"»
probe-diff-api-pr-1.jsonl:18

«"observation": "toolz.itertoolz.pluck.input3"»

| toolz.itertoolz.pluck.input4 | - | same on both sides | same on both sides: {"repr": "<map object>",
"type": "map"}

| probe-diff-api-main-1.jsonl:19 «"observation":
"toolz.itertoolz.pluck.input4"»
probe-diff-api-pr-1.jsonl:19

«"observation": "toolz.itertoolz.pluck.input4"»

| toolz.itertoolz.pluck.input5 | - | same on both sides | same on both sides: {"repr": "<map object>",
"type": "map"}

| probe-diff-api-main-1.jsonl:20 «"observation":
"toolz.itertoolz.pluck.input5"»
probe-diff-api-pr-1.jsonl:20

«"observation": "toolz.itertoolz.pluck.input5"»

| toolz.itertoolz.pluck.input6 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:21 «"observation":
"toolz.itertoolz.pluck.input6"»
probe-diff-api-pr-1.jsonl:21

«"observation": "toolz.itertoolz.pluck.input6"»

| toolz.itertoolz.pluck.input7 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:22 «"observation": "toolz.itertoolz.pluck.input7"»
probe-diff-api-pr-1.jsonl:22

«"observation": "toolz.itertoolz.pluck.input7"»

| toolz.itertoolz.pluck.input8 | - | probe could not run | same on both sides: {"raises": "TypeError"} |

probe-diff-api-main-1.jsonl:23 «"observation": "toolz.itertoolz.pluck.input8"»
probe-diff-api-pr-1.jsonl:23

«"observation": "toolz.itertoolz.pluck.input8"»

| toolz.itertoolz.pluck.input9 | - | same on both sides | same on both sides: {"repr": "<map object>", "type": "map"}

| probe-diff-api-main-1.jsonl:24 «"observation": "toolz.itertoolz.pluck.input9"»
probe-diff-api-pr-1.jsonl:24

«"observation": "toolz.itertoolz.pluck.input9"»

Citation check (PASS): 68 citation(s), every one re-read from its log file and matched.

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (2.3 s run).

Coverage gate (python, confidence normal): 3 of 3 changed executable line(s) ran (100%).

| Changed file | Changed executable lines | Of those, ran

| toolz/itertoolz.py | 3 | 3

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested 508f44f669c7 of the PR against main (source-pair route).

• 3 alternating runs per side: main 3, PR 3 of 3.

• pytest exit codes explained: all runs exited 0.

• Changed code was exercised (coverage gate): 3 of 3 changed executable line(s) ran (100%).

• Probes ran on both sides: 3 probe(s), 3 alternating runs per side, every run printed its observations.

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every claim cites a real log line: 6 citation(s), every one re-read from its log file and matched.

• Every probe observation cites its output line: 68 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

Nothing went wrong.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | PASS | main 3, PR 3 of 3

| pytest exit codes explained | PASS | all runs exited 0

| Changed code was exercised (coverage gate) | PASS | 3 of 3 changed executable line(s) ran (100%)

| Probes ran on both sides | PASS | 3 probe(s), 3 alternating runs per side, every run printed its observations

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

| Nothing posted upstream | PASS | GitHub was only read

| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

| Every claim cites a real log line | PASS | 6 citation(s), every one re-read from its log file and matched

| Every probe observation cites its output line | PASS | 68 citation(s), every one re-read from its log file and

matched

Checklist

• I read the verdict and the plain-English summary.

• Every 'false' or 'unmentioned change' row has a log line I can open.

• Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).

• I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide

REPORT=poc-main/reports/daily/2026-11-14/toolz-665 DECISION=approve (or reject,

approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/pytoolz/toolz/pull/665

Evidence links

• Pull request pytoolz/toolz#665
• record.json
• pr.diff
• setup-main.log
• setup-pr.log
• git.log
• pytest-main-1.log
• pytest-pr-1.log
• pytest-main-2.log
• pytest-pr-2.log
• pytest-main-3.log
• pytest-pr-3.log
• junit-main-1.xml
• junit-pr-1.xml
• coverage-pytest.log
• coverage-pr.json
• probe-model-probe1-main-1.jsonl

• probe-model-probe1-pr-1.jsonl
• probe-model-probe2-main-1.jsonl
• probe-model-probe2-pr-1.jsonl
• probe-diff-api-main-1.jsonl
• probe-diff-api-pr-1.jsonl
• main run 1 result: pytest-main-1.log:8 «78 passed in 0.88s»
• pr run 1 result: pytest-pr-1.log:8 «80 passed in 0.99s»

Glossary

| Term | Meaning
| main | The library as it is today, without the PR.
| PR | Pull request: a proposed change to the library.
| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.
| flaky | A test that gives different results when nothing changed; it is ignored as evidence.
| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-14 12:59 IST